Translation lattices and marked finite quotients

Human Compression Project

1 Introduction

Many finite polyhedra in the companion paper come from one of two infinite affine groups. The parameter changes a lattice of translations, while the finite linear group and the distinguished dihedral subgroups stay fixed. Consequently, the proof of the intersection property can be reduced to a finite list of translation differences and then applied to every period at once.

Two issues must be separated. An affine model satisfying a presentation is initially only a quotient of the presented group. We first identify its full translation kernel, using an exact subgroup presentation. We then determine which period quotients preserve the marked subgroup intersections. The compatible-product construction handles a different operation: changing adjacent orders while retaining an already understood marked quotient.

We include complete proofs of the kernel, deletion and square-group statements used in that construction. They are reused from the project's accepted Topic A theory, with their labels retained for traceability; they are paid parts of this independent common paper. No theorem about forest markings, nilpotent enumeration, or arbitrary coset geometries is silently invoked. The affine arguments reorganize the abelian-normal-subgroup method already used by Hou–Feng–Leemans. They replace separate seed calculations by a common point quotient and a single kernel presentation for each affine group. This is a statement about proof reuse, not historical priority.

2 Rank-three markings

Throughout, $x^g = g^{-1}xg$ and $[x, y] = x^{-1}y^{-1}xy$.

A rank-three involutory marking is an ordered generating tuple of nonidentity involutions. For a subset I of its indices, G_I denotes the subgroup on those markers. The intersection property is $G_I \cap G_J = G_{I\cap J}$. We retain the standard regular-polytope correspondence. The following standard rank-three criterion is included with its elementary proof.

Rank-three test. For three distinct involutions with commuting endpoints and adjacent products of order at least three, IP is equivalent to $G_{01} \cap G_{12} = G_1$.

Proof. Necessity is immediate. For sufficiency, the endpoint subgroup is $G_{02} = \{1, r_0, r_2, r_0r_2\}$. If $G_{02} \cap G_{01}$ contains r_2 or r_0r_2, then $r_2 \in G_{01} \cap G_{12} = G_1$, contradicting distinctness. Thus this intersection is G_0; dually $G_{02} \cap G_{12} = G_2$. These equalities exclude every generator from the subgroup on the other two. Intersections involving singletons now follow, and intersections involving the empty or full index set are immediate. This covers all subset pairs. □

3 Marked intersections and supported extensions

A marked group is a group G together with a finite indexed generating family $(r_i)_{i\in I}$. Except where stated, the markers need not be involutions or distinct. For $S \subseteq I$, put

$$G_S = \langle r_i : i \in S\rangle, \qquad G_\varnothing = 1.$$

We call these subgroups parabolics, retaining their index sets even when two of them happen to coincide. The marking has the intersection property, or IP, if

$$G_S \cap G_T = G_{S\cap T} \quad (S, T \subseteq I). \tag{M1}$$

A marked homomorphism sends each source marker to the corresponding target marker. Its image on G_S is therefore the target parabolic with index set S.

Lemma UA.K0 (lifting one intersection). Let $q : H \to Q$ be a homomorphism and let $C \leq A \cap B$, where $A, B, C \leq H$. If

$$q(A) \cap q(B) = q(C),$$

then

$$A \cap B = C \quad \Longleftrightarrow \quad A \cap B \cap \ker q \leq C. \tag{M2}$$

Proof. The forward implication is immediate. For the converse, let $x \in A \cap B$. Its image belongs to $q(C)$, so choose $c \in C$ with $q(c) = q(x)$. Then $xc^{-1} \in A \cap B \cap \ker q$, which is contained in C by hypothesis. Multiplying by c gives $x \in C$. The inclusion $C \leq A \cap B$ gives equality. □

Theorem UA.K (kernel profiles). Let $q : H \twoheadrightarrow Q$ be a marked epimorphism and suppose that the marking of Q has IP. Write $K = \ker q$ and $K_S = K \cap H_S$. Then H has IP if and only if

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I). \tag{M3}$$

Proof. Apply UA.K0 to $A = H_S$, $B = H_T$, and $C = H_{S\cap T}$. The image hypothesis is exactly IP in Q. The subgroup on the left of the containment in (M2) is $K_S \cap K_T$. Its containment in $H_{S\cap T}$ is equivalent to its containment in $K_{S\cap T}$; the reverse inclusion always holds. This proves each required intersection and its converse. □

A deletion for $S \subseteq I$ is an endomorphism $e_S : G \to G$ satisfying $e_S(r_i) = r_i$ for $i \in S$ and $e_S(r_i) = 1$ otherwise. Requiring these values matters: a general retraction onto G_S need not be a deletion.

Proposition UA.D (Boolean deletions). If a marked group has a deletion e_S for every $S \subseteq I$, then

$$e_S e_T = e_{S\cap T}, \qquad \operatorname{im} e_S = \operatorname{Fix}(e_S) = G_S. \tag{D1}$$

The marking has IP, and

$$\ker e_S = \langle\!\langle r_i : i \notin S \rangle\!\rangle_G, \qquad G = (\ker e_S) \rtimes G_S. \tag{D2}$$

Proof. The two maps in the first equality agree on every generator. The image is generated by precisely the retained markers, and each of those markers is fixed; this proves the image and fixed-point statements. An element of $G_S \cap G_T$ is fixed by $e_S e_T = e_{S\cap T}$, so belongs to $G_{S\cap T}$. The reverse inclusion is immediate.

Let M be the normal closure in (D2). Clearly $M \leq \ker e_S$. In G/M, every omitted marker is trivial, so the image of any word equals the image of its deletion. A word in $\ker e_S$ is therefore trivial in G/M, proving equality. Finally, for $g \in G$, the factorization $g = (g e_S(g)^{-1}) e_S(g)$ has factors in $\ker e_S$ and G_S. Their intersection is trivial because e_S fixes G_S. This is the asserted semidirect product. □

We now assemble extensions whose extra kernel is invisible on each proper part of its support. The groups and kernels may be infinite and nonabelian.

Theorem UA.SK (supported-kernel realization). Let I be finite and let $\mathcal{F}$ be a finite family of distinct subsets of I. Let $B = \langle b_i : i \in I \rangle$ have IP. For every $S \in \mathcal{F}$, suppose the following data are given:

1. A deletion homomorphism $p_S : B \to B_S$, fixing b_i for $i \in S$ and killing b_i otherwise.
2. A group $E_S = \langle e_{S,i} : i \in S \rangle$ and a marked epimorphism $q_S : E_S \to B_S$.
3. Injectivity of q_S on $(E_S)_T = \langle e_{S,i} : i \in T \rangle$ for every $T \subsetneq S$.

Put $N_S = \ker q_S$, and form the group

$$H = \left\{ (b, (x_S)_{S \in \mathcal{F}}) \in B \times \prod_{S \in \mathcal{F}} E_S \,:\, q_S(x_S) = p_S(b) \text{ for every } S \right\}. \tag{K1}$$

For $i \in I$, define h_i to have base coordinate b_i and S-coordinate $e_{S,i}$ when $i \in S$, and 1 otherwise. These elements generate all of H. If $\pi : H \to B$ is the base projection and $H_L = \langle h_i : i \in L \rangle$, then, for every $L \subseteq I$,

$$\pi(H_L) = B_L, \qquad \ker \pi = \prod_{S \in \mathcal{F}} N_S, \qquad \ker \pi \cap H_L = \prod_{\substack{S \in \mathcal{F} \\ S \subseteq L}} N_S. \tag{K2}$$

The factors on the right are embedded as pure coordinate subgroups. The marking of H has IP. If B and all E_S are finite, then

$$|H_L| = |B_L| \prod_{\substack{S \in \mathcal{F} \\ S \subseteq L}} |N_S|. \tag{K3}$$

Proof. The compatibility equations in (K1) are preserved by products and inverses, so H is a subgroup of the displayed direct product. The prescribed tuple h_i satisfies those equations by the literal deletion rule. Let H' be the subgroup generated by these tuples. Its base image is B, and the projection of $H'_S = \langle h_i : i \in S \rangle$ onto E_S is onto.

We first prove, by induction on inclusion among the finitely many supports, that every pure coordinate element of N_S belongs to H'_S. Fix $n \in N_S$. Choose a word on the markers indexed by S which evaluates to n in E_S, and evaluate the same word in H', obtaining $w \in H'_S$. Its base coordinate lies in B_S and is $q_S(n) = 1$.

Consider its coordinate at another support T. Because the word uses only markers from S, that coordinate lies in $(E_T)_{S \cap T}$. Its image under q_T is $p_T(1) = 1$. If $T \not\subseteq S$, then $S \cap T \subsetneq T$, and the proper-parabolic injectivity hypothesis forces this coordinate to be 1. The only possible nonidentity coordinates of w are consequently at supports $T \subseteq S$. Its S-coordinate is the prescribed n, while every coordinate at a proper support $T \subsetneq S$ belongs to N_T.

For each such proper support, the induction hypothesis supplies the pure coordinate element having exactly that value in N_T, as an element of $H'_T \leq H'_S$. Multiply w by the inverses of those elements. Distinct pure coordinate subgroups commute in the ambient direct product, so these operations remove the unwanted coordinates without changing the S-coordinate. The result is the pure element n in coordinate S, and remains in H'_S. A minimal support has no smaller coordinates to remove, so this argument includes the initial step. No commutativity within an individual kernel has been used.

The kernel of π in the full group H is exactly the direct product of the N_S, by (K1). The induction places that entire kernel in H'. Since H' already maps onto B, every $h \in H$ has the same base coordinate as some $h' \in H'$; then $h(h')^{-1} \in \ker \pi \leq H'$. Thus $H' = H$.

The image equality $\pi(H_L) = B_L$ follows on generators. If $x \in H_L \cap \ker \pi$, its S-coordinate lies in $(E_S)_{L \cap S}$ and in N_S. This coordinate is trivial whenever $S \not\subseteq L$, again by proper-parabolic injectivity. Conversely, when $S \subseteq L$, the isolation argument placed the entire pure N_S in $H_S \leq H_L$. This proves all three assertions in (K2).

The coordinate products in (K2) satisfy

$$\left(\prod_{S \subseteq L} N_S \right) \cap \left(\prod_{S \subseteq M} N_S \right) = \prod_{S \subseteq L \cap M} N_S.$$

Theorem UA.K now proves IP in H. More explicitly, an element of $H_L \cap H_M$ has base in $B_{L \cap M}$; subtracting a lift from $H_{L \cap M}$ leaves an element in both displayed kernel products, hence in the kernel product for $L \cap M$. This also shows exactly where IP of the base is used. Finally (K3) is the order formula for the surjection $H_L \to B_L$ with the kernel just computed. □

Corollary UA.SK.Local (recognition on an antichain). Under UA.SK, if $\mathcal{F}$ is an antichain under inclusion, then the coordinate projection $H_S \to E_S$ is a marked isomorphism for every $S \in \mathcal{F}$.

Proof. It is onto on generators. If an element maps to 1, its base coordinate also vanishes, since the base value of a word on S is the q_S-image of its S-coordinate. Formula (K2) says that the remaining kernel in H_S has only the factor N_S: the antichain excludes every other support contained in S. The S-coordinate detects that factor, so the element is 1. □

Corollary UA.SK.Delete (derived deletions). If, in addition, B has deletions for every subset of I, then every E_S and H has all its Boolean deletions.

Proof. For $T \subsetneq S$, the restriction $q_S : (E_S)_T \to B_T$ is a marked isomorphism: it is onto on generators and injective by hypothesis. Compose q_S, the deletion $B_S \to B_T$, and the inverse of this isomorphism. This gives a homomorphism $E_S \to (E_S)_T$ that fixes exactly the T-markers and kills the others. For $T = S$ use the identity. Proposition UA.D supplies the Boolean identities.

For $L \subseteq I$, apply the base deletion e_L and, in coordinate S, the local deletion onto $S \cap L$. The two sides of each compatibility equation agree on the local markers; the base deletions commute by UA.D. Thus the coordinatewise operation preserves (K1). On h_i it fixes h_i if $i \in L$, and kills it otherwise. It is the required deletion of H. □

4 Square presentations and their translation periods

Let

$$a(x, y) = (-x, y), \quad b(x, y) = (y, x), \quad c(x, y) = (x, 1 - y).$$

Proposition UA.Affine (the complete square presentation). These generate $W = \mathbb{Z}^2 \rtimes D_8$, of type $\{4, 4\}$, with $ac = ca$. Its presentation is

$$W = \langle a, b, c \mid a^2 = b^2 = c^2 = (ab)^4 = (bc)^4 = (ac)^2 = 1 \rangle. \tag{S1}$$

Proof. Put $d = bab$, $y_0 = cd$, and $x_0 = by_0b$. The relations give $[a, d] = 1$, $[a, y_0] = 1$, $dy_0d = y_0^{-1}$, $ax_0a = x_0^{-1}$, and conjugation by b swaps x_0, y_0. Substitution of $bc = x_0ab$ makes $(bc)^4 = 1$ equivalent to $[x_0, y_0] = 1$. Thus every word collects to $x_0^u y_0^v M$, $M \in \langle a, b \rangle = D_8$. In the affine model x_0, y_0 are the unit translations, so these forms are distinct. This proves (S1). □

The normal form also fixes our affine multiplication convention: the rightmost transformation acts first. The symbol D_8 denotes the group of signed permutation matrices in two coordinates, of order eight. In all dihedral notation below the subscript denotes the order of the group.

Proposition UA.Square (finite square seeds). For $n \geq 5$, define the invariant lattice

$$\Lambda_n = \begin{cases} 2^e\mathbb{Z}^2, & n = 2e + 3, \\ 2^e\langle (1, 1), (1, -1) \rangle, & n = 2e + 4, \end{cases} \qquad e \geq 1,$$

and put $H_n = W/\Lambda_n$. Its order is 2^n; the full linear D_8 coordinate is retained, even if its action on the vertex quotient is not faithful. Its pair orders are $4, 4, 2$. It has all Boolean deletions and IP. Its third lower-central subgroup is $2\mathbb{Z}^2/\Lambda_n$, of order 2^{n-5}.

Proof. The lattice indices are respectively 2^{2e} and 2^{2e+1}, so the normal form gives the stated orders. The three linear pair products have orders four, four and two; hence the marked pair orders cannot fall in either lattice quotient.

Write $T = abcb$, $U = cbab$, translations by $(-1, 0)$, $(0, 1)$. The extra relators for H_n can be taken as T^{2^e}, U^{2^e} in the square case, and $T^{2^e}U^{2^e}, T^{2^e}U^{-2^e}$ in the diagonal case. After deleting any marker, both T, U have order at most two: deleting a leaves conjugates of c, deleting c leaves conjugates of a, and deleting b leaves ac, ca. Therefore every lattice relator survives every deletion. The even Coxeter relators also survive. Thus H_n has Boolean deletions and IP by UA.D.

Moreover,

$$\gamma_3(W) = 2\mathbb{Z}^2, \qquad \gamma_3(H_n) = 2\mathbb{Z}^2/\Lambda_n, \qquad |\gamma_3(H_n)| = 2^{n-5}. \tag{S2}$$

Indeed reduction modulo $2\mathbb{Z}^2$ makes the adjacent squares central; conversely $[[a,b],c]$ and $[[b,c],a]$ give the two doubled unit translations, up to signs. Thus H_5 is class two of order 32. The squared translations are not always independent cyclic subgroups: in H_6 they coincide and have order two. Formula (S2) uses a lattice index and does not require independence of those cyclic subgroups. □

5 Recognizing an affine presentation

We write an integral affine transformation as (M,t), acting by $v \mapsto Mv + t$, with multiplication

$$(M,t)(N,u) = (MN, t + Mu). \tag{1}$$

Let E be a group generated by finitely many such transformations, and let $Q \le \mathrm{GL}_d(\mathbb{Z})$ be its finite linear image. Its translation kernel L is a subgroup of $\mathbb{Z}^d$, invariant under Q. A translation is written $t_v = (I, v)$.

Lemma AL.1 (fixed Schreier lattice). Choose a word lift $\hat{q} \in E$ for each $q \in Q$, with $\hat{1} = 1$. Then L is generated by the translations

$$\widehat{\bar{s}q}^{\,-1}\, s\hat{q}, \qquad s \text{ a distinguished generator}, \quad q \in Q. \tag{2}$$

Proof. The products have trivial linear image. Write a partial word as $\hat{q}n$, with $n \in L$. Left multiplication by s changes this to $\widehat{\bar{s}qt}n$, where t is the transition in (2). Induction expresses a word of trivial linear image as a product of these transitions; its final representative is $\hat{1} = 1$. Inverses introduce no extra generator: the transition for s^{-1} at q is the inverse of the transition for s at $\bar{s}^{-1}q$. Thus the induction applies to words with positive and negative letters. Every transition itself is in L, so they generate exactly L. □

The normalization in (2) matters. The raw differences $s\hat{q}\widehat{\bar{s}q}^{\,-1}$ need not generate L without conjugation. Each is a conjugate of the corresponding transition by $\widehat{\bar{s}q}$. Consequently their Q-orbits generate L. In particular, if a proposed Q-invariant lattice contains every raw difference, it contains every normalized transition and hence L. If exhibited raw differences also span the proposed lattice, equality follows. This invariant-lattice version justifies the raw-difference format of the unchanged earlier $\{6,6\}$ checker [C66]. The new $\{6,4\}$ checker uses (2) directly.

This is a finite proof of a lattice identity. Containment in a proposed lattice is checked on the normalized transitions, or on the raw differences with the stated invariance condition. Equality follows when independent integer combinations give a basis of that lattice, or when the greatest common divisor of the full-rank minors gives its exact index. Sampling finite quotients is not a replacement for this step.

Lemma AL.2 (presentation bridge). Let U be a finitely presented marked group whose relators hold on the affine generators of E. Suppose their linear images define the same epimorphism to Q. If an exact Reidemeister–Schreier presentation proves $\ker(U \to Q) \cong \mathbb{Z}^d$, and the affine translation kernel L has rank d, then the marked epimorphism $U \to E$ is an isomorphism.

Proof. The restriction maps $\ker(U \to Q)$ onto L: an affine translation has a preimage in U, and its linear image is trivial. This is a surjection between free abelian groups of rank d. Tensoring with $\mathbb{Q}$ shows that its kernel has rank zero. A subgroup of a free abelian group is torsion-free, so that kernel is trivial. The remaining kernel of $U \to E$ lies in this subgroup, and is therefore trivial as well. □

The hypothesis concerns the actual kernel presentation. Computing only its abelianization would not establish it: a nonabelian group can have free abelian abelianization of rank d. Our applications retain the exact rewriting and Tietze algorithms used to verify this finite presentation step. It is a computer-assisted input, not a Lean proof.

6 Periods and intersections of distinguished subgroups

Let $w_1, \dots, w_d \in U = E$ be translation words with linearly independent vectors $v_1, \dots, v_d$. Write $\Lambda = \sum_i \mathbb{Z} v_i \le L$ and suppose it is Q-invariant. For a positive integer m, adjoining $w_i^m = 1$ to the complete presentation of U gives exactly

$$E_m = E/(m\Lambda). \tag{3}$$

Indeed, the normal closure of the powers is $m\Lambda$: conjugation by an affine element acts on translations by its linear part, and this lattice is already invariant. Hence

$$|E_m| = |Q|[L : \Lambda]m^d. \tag{4}$$

This proves equality for the presented group, not merely the order of a chosen quotient. If Q is solvable, every E_m is solvable.

Suppose now that the marking has rank three with distinct generators, its endpoints commute, both adjacent products have order at least three, and its two adjacent subgroups $A = E_{01}$ and $B = E_{12}$ are finite and inject into Q. For each $q \in \overline{A} \cap \overline{B}$, let $(q, t_A(q))$ and $(q, t_B(q))$ be their unique affine lifts. Put

$$\Delta(q) = t_A(q) - t_B(q). \tag{5}$$

Theorem AL.3 (period-lattice intersection criterion). The images of A and B in E_m meet precisely in the elements indexed by

$$\{q \in \overline{A} \cap \overline{B} : \Delta(q) \in m\Lambda\}. \tag{6}$$

In particular, if $A \cap B = \langle r_1 \rangle$ and every other value $\Delta(q)$ lies outside Λ, then every E_m, $m \ge 1$, has the full rank-three intersection property and the same adjacent orders as E. Its order is (4).

Proof. Equality of two cosets modulo translations first forces equality of their linear parts. For equal linear part, their affine quotient is translation by the difference (5). This proves (6). Each adjacent subgroup remains injective in the quotient because its linear action is faithful, so its order and the product order of its two reflections are unchanged. The assumed exclusions apply to $m\Lambda \subseteq \Lambda$. Thus the adjacent intersection is exactly the common involution subgroup, and the rank-three test proves all subset intersections. □

The proof separates two possibilities often hidden by equal group-order counts. The point-group intersection may already have order two, as in the $\{6, 4\}$ application. Or it may be larger, with a short list of nonzero translation differences, as in the $\{6, 6\}$ application. Those differences must be excluded from the period lattice explicitly.

One coarse quotient controls all smaller periods

The same argument has a group-theoretic formulation which is useful independently of coordinates. Let A, B, L be subgroups of a group E, with

$$L \cap A = L \cap B = 1.$$

Let $N \trianglelefteq E$ lie in L. If the images of A, B in E/N intersect exactly in the image of $A \cap B$, then the same statement holds in E/M for every normal subgroup $M \le N$.

Indeed, suppose $a \in A$, $b \in B$ have the same image modulo M. They have the same image modulo N, so there is $c \in A \cap B$ with $aN = cN = bN$. Then $ac^{-1} \in N \cap A = 1$ and $bc^{-1} \in N \cap B = 1$, proving $a = b = c$. The reverse inclusion is immediate. In the affine setting L is the translation kernel, $N = \Lambda$ and $M = m\Lambda$. Point-faithfulness supplies precisely the two trivial intersections. This statement, for an arbitrary indexed family of smaller normal periods, is verified in `AffinePeriodIntersections.lean`. The finite coordinate checks which identify L, Λ and the initial overlap remain separate mathematical obligations.

7 Changing one adjacent order

Let $(B; a, b, c)$ be a rank-three string C-group of type $\{h, q\}$, with h, q even, and suppose the map

$$\rho : B \longrightarrow B_{12}, \qquad a \mapsto 1, b \mapsto b, c \mapsto c \tag{7}$$

is a retraction. For k a positive multiple of q, take the standard marked epimorphism $D_{2k} \to B_{12} = D_{2q}$. The compatible product

$$F = \{(g, d) \in B \times D_{2k} : \rho(g) = \overline{d}\} \tag{8}$$

has markers $(a, 1), (b, \tilde{b}), (c, \tilde{c})$. The evenness of h is forced by (7); it also ensures that the first pair has unchanged order h.

Theorem AL.4 (one-edge extension). These markers generate F. If B is finite, its order is $|B|k/q$. Its type is $\{h, k\}$, and it has the intersection property. The projection onto B is injective on the first adjacent subgroup. If B is solvable, so is F.

Proof. The word $(r_1 r_2)^q$ generates the entire cyclic kernel of (8), since its first coordinate is trivial and its second has order k/q. Surjectivity onto B then proves generation and the order formula. The first adjacent subgroup has second coordinate of order at most two, with that coordinate determined by the image under (7); thus its projection is injective. The second adjacent subgroup is the full D_{2k}. If an element belongs to both, its projection belongs to $\langle b \rangle$ by IP in B. Subtract the appropriate common involution. The remaining element lies in the first subgroup and in the projection kernel, hence is trivial. This proves IP by the rank-three criterion. Solvability follows from the cyclic kernel. □

Theorem AL.4 is a specialization of UA.SK; it is included to make clear that only the indicated deletion is needed here. More simultaneous edges use the supported coordinate kernels of that theorem.

There is a useful exact presentation variant. Suppose a complete marked presentation for B contains $(bc)^q = 1$. Replace that relation by $(bc)^k = 1$, retain the remaining relators, and add $[a, (bc)^q] = 1$. Call the resulting presented group H. If every retained relator holds in (8), then $H = F$ as a marked group. Indeed, $C = \langle (bc)^q \rangle$ is normal in H: the last relation centralizes it by a, and b, c invert it. Its order is at most k/q, and $H/C = B$ by the presentation. Consequently $|H| \leq |B|k/q$. The marked surjection onto (8) attains this bound, proving equality and injectivity. The same argument is valid for an infinite base if the finite kernel maps injectively and the induced map on the base is an isomorphism.

Attribution and verification

The affine lattice method, Reidemeister–Schreier rewriting and Tietze transformations are classical. Hou–Feng–Leemans already used free abelian normal subgroups and their power subgroups. Our earlier cubic calculation and supported-kernel theory are incorporated here with their full proofs and exact computational dependencies. The companion application identifies every source word and exceptional parameter.

The regular-polytope correspondence is retained from McMullen–Schulte, *Abstract Regular Polytopes* (2002), Section 2E. The local rank-three intersection test is proved here. The square affine presentation has its elementary normal-form proof here; the cubic presentations require the two complete exact GAP kernel programs. No computation of an abelianization is substituted for a presentation of the actual kernel.

The citation [C66] denotes the project's earlier *One affine model for all three* $\{6, 6\}$ *lattice families*; its complete local proof and two programs are incorporated into the companion application. No additional statement from that note is assumed without its proof here.

The full common paper, the application, and all six mathematical programs belong to the replacement proof. A theorem already available in another topic is not free in an independent comparison. The source's historical programs have not been supplied, so their cost is unknown. No compression factor is asserted in this edition.